

PREDICTING CUSTOMER DISSATISFACTION IN ONLINE RETAIL

An Exploratory and Predictive Task Formulation Using Order-Level Data

Term Paper

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Project basis: Statis_Eval_LM.ipynb was not used for this paper. This paper is based specifically on the online-retail notebook supplied for the study.

Abstract

Customer reviews are an important source of feedback for online retailers because they provide an observable signal of customer satisfaction with a purchase experience. This study investigates whether transaction, payment, product, seller, and order-level information can be organized into a realistic predictive task for identifying orders that are likely to receive low customer review scores. The study is based on a relational online-retail dataset containing interconnected tables for orders, order items, payments, reviews, products, customers, sellers, and product-category translation. The analysis defines a binary target variable, `low_review`, where a review score of 1 or 2 is classified as a low review and scores above 2 are classified as non-low reviews. Only delivered orders are retained because the target concerns post-delivery customer feedback. The tables are then joined and aggregated at the order level to construct predictors such as total price, total freight, number of items, number of sellers, total payment value, payment installments, and representative product category. A major methodological emphasis is prevention of data leakage: review text and review timestamps are excluded because they become available only after the prediction point. Exploratory analysis examines target imbalance and relationships between low reviews and order value, order complexity, payment installments, and product categories. The notebook represents a data preparation and exploratory-analysis stage rather than a completed predictive-modeling experiment; therefore, no model accuracy or classification performance is claimed. The findings indicate that the dataset provides a plausible foundation for a future model that could support proactive customer-service, seller monitoring, and logistics interventions.

Keywords: e-commerce, customer satisfaction, customer reviews, predictive analytics, exploratory data analysis, feature engineering, data leakage, binary classification

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1. Introduction

The growth of online retail has made customer experience a central component of business performance. Unlike traditional retail environments, e-commerce platforms generate detailed digital records of orders, payments, products, sellers, delivery events, and customer feedback. These records create opportunities to use predictive analytics to identify patterns associated with customer dissatisfaction before dissatisfaction is formally expressed.

Customer review scores are particularly useful because they provide a structured measure of the customer's assessment of the purchase experience. However, a review is submitted after the customer has received the order. Consequently, a predictive system intended to intervene before a review is submitted must use only information available before or at the prediction point. This temporal distinction is fundamental to the design of a valid predictive task.

The supplied notebook formulates the problem as predicting whether a delivered order will receive a low customer review score of two stars or fewer. The analysis does not immediately train a machine-learning model. Instead, it first establishes the predictive setting, examines the relational data structure, creates an order-level analytical table, removes post-outcome information, handles missing values, and performs task-driven exploratory analysis. This sequence is appropriate because a predictive model is only meaningful when the target definition, prediction timing, feature availability, and data quality have been established.

The study therefore focuses on a foundational question: can the available online-retail transaction data be transformed into a realistic and useful prediction problem for customer dissatisfaction? The answer is evaluated through systematic data preparation and exploratory analysis rather than through model-performance metrics.

2. Background and Related Concepts

2.1 Predictive Analytics in Online Retail

Predictive analytics uses historical observations to identify patterns that can support predictions about future or currently unknown outcomes. In online retail, predictive approaches can be applied to sales forecasting, customer behavior, delivery performance, churn, fraud, and customer satisfaction. The value of prediction is not limited to classification accuracy; it depends on whether predictions can support timely operational decisions.

2.2 Customer Reviews as a Satisfaction Signal

Customer reviews provide a direct post-purchase assessment of the customer's experience. The online-retail dataset used in this project contains review scores ranging from one to five, allowing the outcome to be transformed into a binary dissatisfaction indicator. The notebook defines scores of one or two as low reviews. This threshold focuses the analysis on clearly negative experiences rather than treating neutral ratings as dissatisfaction.

2.3 Exploratory Data Analysis

Exploratory Data Analysis (EDA) is used to understand distributions, relationships, anomalies, and potential predictive signals before formal modelling. The notebook applies count plots and boxplots to examine the target distribution and relationships between low reviews and order-level variables. This approach is consistent with the broader purpose of EDA: discovering structure in data before selecting or fitting predictive models.

2.4 Data Leakage and Prediction Timing

Data leakage occurs when information that would not be available at the intended prediction time is inadvertently used as a predictor. This can produce apparently strong model performance that does not generalize to real deployment. The notebook explicitly addresses this risk by removing review comments, review creation dates, and review-answer timestamps. This is especially important in the present problem because these variables are directly connected to the outcome and are only available after the customer-review process has occurred.

3. Research Problem, Objectives and Questions

3.1 Research Problem

Online retailers need to identify orders that may result in customer dissatisfaction early enough to take corrective action. A review score is only observed after delivery, so using the review itself to predict the review would not provide operational value. The research problem is therefore to determine whether information available by the time an order is handed to the carrier can support the identification of orders likely to receive a low review score.

3.2 Research Objective

The primary objective is to develop and evaluate a realistic data foundation for predicting whether a delivered online-retail order will receive a low customer review score (≤ 2 stars).

3.3 Specific Objectives

- Define a business-relevant binary prediction target for customer dissatisfaction.
- Identify the information available at the intended prediction point.
- Integrate the relational order, item, payment, product, and review data into an order-level analytical dataset.
- Create meaningful aggregated predictors from one-to-many transactional tables.
- Remove variables that would cause post-outcome information leakage.
- Handle missing numerical and categorical values.
- Explore class balance and relationships between low reviews and selected operational variables.
- Identify limitations and requirements for a future predictive modelling stage.

3.4 Research Questions

- Can online-retail transaction data be formulated into a realistic prediction task for low customer review scores?
- Which order, payment, product, and seller-related variables can be constructed before the review outcome is known?
- How does order-level aggregation affect the representation of item-level and payment-level information?

- What exploratory patterns are visible between low reviews and order characteristics such as price, number of items, installments, and product category?
- What risks and limitations must be addressed before deploying a predictive model?

4. Dataset and Data Architecture

The notebook loads eight interconnected CSV tables: `orders.csv`, `order_items.csv`, `order_payments.csv`, `order_review.csv`, `products.csv`, `product_category_name_translation.csv`, `customers.csv`, and `sellers.csv`. The structure matches the Brazilian E-Commerce Public Dataset by Olist, a public dataset containing approximately 100,000 orders and information covering order status, products, payments, freight, customers, sellers, and reviews. The original dataset is organized relationally rather than as a single flat table.

The `orders` table is treated as the central transaction entity. The `order_items` table may contain multiple records for a single order, while `order_payments` can also contain multiple payment records. The `review` table records post-delivery customer feedback. Products provide product attributes and categories, while customers and sellers provide contextual information.

Table	Role in the analysis	Important information
<code>orders</code>	Central order entity	Status, purchase and delivery-related timestamps
<code>order_items</code>	Item-level detail	Product, seller, price and freight
<code>order_payments</code>	Payment detail	Payment value and installments
<code>order_review</code>	Outcome source	Review score and post-delivery review information
<code>products</code>	Product context	Product category and physical attributes
<code>product_category_name_translation</code>	Category translation	Portuguese-to-English category mapping

customers	Customer context	Customer identification and location
sellers	Seller context	Seller identification and location

5. Research Methodology

5.1 Research Design

The notebook follows a structured predictive-analytics preparation pipeline. It begins with task definition and prediction timing, proceeds through data loading and structural inspection, filters the population to delivered orders, creates the target variable, aggregates one-to-many data to the order level, removes leakage-prone variables, handles missing values, and finally conducts exploratory analysis.

5.2 Prediction Timing

The assumed prediction point is when the order is handed over to the carrier for delivery. At this point, order information, payment information, product characteristics, seller information, and estimated delivery information can be available. The actual review outcome is not yet known.

5.3 Target Construction

The target variable is `low_review`. The notebook assigns a value of 1 when `review_score` is less than or equal to two, and 0 otherwise. This converts the original five-level review score into a binary classification problem centered on clear dissatisfaction.

5.4 Population Filtering

Only orders with `order_status` equal to `delivered` are retained. This is logically consistent with the target because undelivered or cancelled orders cannot be interpreted in the same way as completed customer experiences that can receive a review.

6. Data Preparation and Feature Engineering

6.1 Order-Level Item Aggregation

Because an order can contain multiple items, the notebook groups `order_items` by `order_id`. Four order-level variables are constructed: `total_price`, `total_freight`, `num_items`, and `num_sellers`. Total price is the sum of item prices, total freight is the sum of shipping charges, `num_items` counts products, and `num_sellers` counts distinct sellers involved in the order.

6.2 Payment Aggregation

Payment records are also aggregated by order. The notebook calculates total payment value and retains the maximum number of payment installments. These variables represent the financial structure of the purchase and may provide information relevant to customer experience.

6.3 Product Category Feature

Product information is merged with the category-translation table. Because an order may contain several products and therefore several categories, the most frequent English product category is selected as a representative order-level category. This is a practical simplification that converts item-level categorical information into an order-level feature.

6.4 Leakage Prevention

The notebook removes `review_comment_message`, `review_creation_date`, and `review_answer_timestamp`. These variables are post-delivery and therefore unavailable at the intended prediction point. Keeping them would allow the model to learn information that would only be known after the customer had already expressed dissatisfaction.

6.5 Missing-Value Handling

Numerical columns are intended to be filled using their column medians, while remaining missing values are replaced with the categorical label 'Unknown'. Median imputation is relatively robust to extreme values, while the Unknown category preserves observations instead of deleting rows.

7. Exploratory Data Analysis

7.1 Target Distribution

The notebook visualizes the distribution of `low_review` using a count plot. The analysis identifies low reviews as a minority class. This indicates that any future classification model should account for class imbalance and should not rely on accuracy alone.

7.2 Total Price and Low Reviews

A boxplot compares `total_price` between low-review and non-low-review orders. The purpose is to determine whether order value differs systematically between the two outcome groups. Such a relationship could make order value useful as a predictive feature, although a visual difference alone cannot establish causality.

7.3 Number of Items and Low Reviews

The notebook compares `num_items` across the two review groups. Order complexity may affect customer experience because larger or more complex orders can involve more products, sellers, and logistical interactions. The visualization therefore acts as an initial feasibility check for this feature.

7.4 Payment Installments and Low Reviews

Payment installments are compared between low-review and other orders. Installment behavior may reflect purchasing patterns and financial commitment. However, the analysis should be interpreted as an association rather than evidence that installment choice causes dissatisfaction.

7.5 Product Categories and Low Reviews

The notebook visualizes the ten most frequent product categories and separates their counts by low-review status. This analysis can reveal whether dissatisfaction is concentrated in particular product categories. Such patterns could later support category-level modelling, seller monitoring, or quality-control investigations.

8. Findings and Discussion

The notebook establishes that the online-retail data can support a coherent prediction problem, provided that the prediction point is explicitly defined and post-outcome variables are excluded.

The relational nature of the dataset is an important methodological consideration: meaningful predictors are not immediately available in a single table and must be constructed through joins and aggregations.

The target definition focuses on clearly negative experiences by classifying one- and two-star reviews as low reviews. This produces a business-oriented outcome that could be useful for proactive intervention. The exploratory analysis also indicates that low reviews form a minority class, creating a class-imbalance issue for any subsequent classifier.

The engineered variables provide several dimensions of customer experience. Monetary variables represent order value and shipping cost; structural variables represent order complexity; payment variables represent payment behavior; and product-category variables represent the type of product purchased. Together, these variables create a richer representation than the original order table alone.

An important contribution of the notebook is its explicit attention to leakage. Removing review text and review timestamps prevents the analysis from using information generated after the outcome. This makes the proposed prediction setting more realistic and improves the validity of any future model evaluation.

At the same time, the exploratory stage cannot establish which variables will be most predictive. The boxplots and category counts are descriptive tools rather than inferential tests. A future study should therefore use a carefully designed train-validation-test procedure, appropriate imbalance handling, and classification metrics such as precision, recall, F1-score, ROC-AUC and, where appropriate for an imbalanced target, precision-recall AUC.

9. Limitations, Risks and Assumptions

- Review scores are subjective and may reflect individual expectations rather than objective product or service quality.
- Only customers who submit reviews are represented in the target outcome, which can create self-selection bias.
- Item-level information is aggregated to the order level, potentially hiding dissatisfaction associated with a specific product within a multi-item order.
- The representative product category is chosen using the mode, which simplifies orders containing several categories.

- The dataset does not capture every operational factor that may affect satisfaction, such as internal logistics decisions or customer-service interactions.
- The assumed prediction point at carrier handover simplifies real-world operational timing.
- The current notebook performs preparation and exploratory analysis but does not train or evaluate a predictive model; therefore, predictive performance cannot yet be concluded.
- Class imbalance may cause misleadingly high accuracy if a future model favors the majority class.

10. Recommendations and Future Work

The next stage should convert the prepared order-level dataset into a complete supervised-learning pipeline. The dataset should first be split into training and testing data using a strategy consistent with the temporal nature of the problem. Feature transformations should be learned from the training data only.

- Develop baseline models such as Logistic Regression, Decision Tree, Random Forest and Gradient Boosting.
- Compare models using precision, recall, F1-score, ROC-AUC and precision-recall AUC rather than accuracy alone.
- Investigate class-weighting or resampling methods because low reviews are a minority outcome.
- Add delivery-performance variables that are available before or at carrier handover, such as estimated delivery timing and other legitimate operational features.
- Evaluate whether seller-level and product-category-level features improve prediction while avoiding information that becomes available after the outcome.
- Use temporal validation where possible so that future orders are not predicted using information from later periods.
- Perform feature importance or explainability analysis to identify actionable drivers of predicted dissatisfaction.
- Translate model outputs into operational risk bands that can support customer-service or logistics interventions.

11. Conclusion

This study formulated a realistic predictive-analytics problem for online retail: identifying delivered orders that are likely to receive low customer review scores. Based on the supplied notebook, the problem is framed around a prediction point at carrier handover, ensuring that the model would rely only on information that could be known before the customer submits a review.

The analysis integrates multiple relational tables into an order-level representation and creates meaningful predictors from item, payment, and product information. It also removes review information that would create data leakage and applies missing-value handling before exploratory analysis. The resulting analytical structure provides a reasonable foundation for a subsequent machine-learning classification stage.

The main conclusion is therefore not that a particular machine-learning algorithm has already achieved a specific level of accuracy. The current notebook does not contain a completed modelling experiment. Instead, it demonstrates that the data can be transformed into a coherent and business-relevant prediction task, while also identifying important risks such as class imbalance, selection bias, aggregation loss, and temporal leakage.

A future modelling stage can build on this foundation to determine whether low-review orders can be predicted with sufficient reliability to support proactive customer-service, seller-quality monitoring, and logistics improvements.

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Appendix A: Notebook-to-Paper Traceability

Notebook component	Paper section
Notebook task definition	Sections 1 and 3
Eight-table data loading	Section 4
Delivered-order filtering	Sections 5 and 6
low_review target construction	Sections 3 and 5
Item aggregation	Section 6.1
Payment aggregation	Section 6.2
Product category aggregation	Section 6.3
Leakage removal	Section 6.4
Missing-value handling	Section 6.5
Target distribution	Section 7.1
Price, item-count and installment analysis	Sections 7.2–7.4
Product-category analysis	Section 7.5
Limitations and assumptions	Section 9